MENTAL HEALTH MONITORING SYSTEM

Phase 3 — End-to-End Three-Head Classifier Training

Progress Report

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Platform: Kaggle Notebooks | Date: July 18, 2026

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1. Project Overview

The Mental Health Monitoring System is a multi-phase NLP project designed to automatically detect depression, classify mental health severity, and generate clinically coherent reports from raw text input. The system is structured as a four-layer architecture:

- Layer 1 — Transformer backbone (MentalRoBERTa) for contextual text embeddings
- Layer 2 — Graph Attention Network (GAT) for structural reasoning over token dependency graphs
- Layer 3 — Three-head classifier: emotion recognition, mental health category, and severity regression
- Layer 4 — Large Language Model (Groq) for report generation and crisis alerting

Phase 3 is the first phase where all layers run together: the Phase 1 backbone, the Phase 2 GAT, and three classification heads are connected into a single pipeline and trained jointly, rather than each layer being trained in isolation on top of a frozen predecessor.

2. Phase 3 Objective

Connect the MentalRoBERTa backbone, the GAT, and three classification heads (emotion, mental health category, severity) into one end-to-end model, and jointly fine-tune the backbone and GAT together with the emotion and MH-category heads. The severity head is included in the architecture but kept frozen, since PHQ-8 severity training requires the DAIC-WOZ dataset, which is still pending a data use agreement.

2.1 Deviations from the Original Plan

Three aspects of this notebook depart from the original Phase 3 plan; the first is flagged directly in the notebook itself, the other two are worth calling out explicitly:

- **Dataset substitution:** DepressionEmo was the originally planned training set for the emotion head. It was unavailable via HuggingFace at the time of training, so GoEmotions — the same dataset already used in Phase 1, Stage 2 — was used in its place. The emotion head in Phase 3 is therefore trained on data the backbone has already seen once during Phase 1 fine-tuning, rather than on a new, independent emotion-labelled corpus.
- **Edge construction:** Phase 2's GAT was trained on spaCy dependency-parse edges (grammatical child→head relationships). For end-to-end training here, the GAT instead uses simple sequential edges (token i → token i+1), pre-built and cached per (batch size, sequence length) pair. This is a reasonable engineering simplification — dependency parsing is not differentiable and would be slow to run per batch inside a training loop — but it means the GAT is no longer operating on the same graph structure it was trained on in Phase 2. That is worth keeping in mind when interpreting how much the transferred Phase 2 weights actually contribute here.
- **MH category labels are synthetic:** there is no human-annotated depressed / non-depressed label in GoEmotions. A comment is labelled "depressed" if any of the 28 emotion labels is active at all. This is a weak proxy that does not specifically capture depression-relevant emotion, so the MH-category head's results should be read with that limitation in mind.

3. Environment and Configuration

Platform	Kaggle Notebooks

Hardware	GPU (CUDA)
Frameworks	PyTorch, PyTorch Geometric (GATConv), HuggingFace Transformers, datasets, accelerate
Phase 1 checkpoint	mentalroberta_phase1_final — located dynamically via glob, found successfully
Phase 2 checkpoint	gat_phase2_best.pt — located dynamically via glob, found successfully
Max Sequence Length	128 tokens

4. Dataset — GoEmotions

4.1 Loading and Cleaning

Source	3 raw CSV files (goemotions_1/2/3.csv), combined into one dataframe
Raw shape	211,225 rows × 37 columns (one row per annotator rating, plus metadata)
Unclear examples removed	Rows where example_very_unclear is True are dropped before aggregation
Aggregation	groupby("id"), max() across raters per emotion — an emotion is present if any rater flagged it
After deduplication	58,009 unique comments × 28 emotion labels — matches the official GoEmotions paper's reported size exactly

4.2 Train / Validation / Test Split

Split	Size	Proportion
Train	49,307	85%
Validation	4,351	7.5%
Test	4,351	7.5%

random_state = 42 makes the split reproducible across sessions. The MentalRoBERTa tokenizer from Phase 1 is reused, with a fixed max length of 128 tokens (padding + truncation). Batch sizes are 16 for training and 32 for validation/testing.

5. Phase 3 Model Architecture

The Phase3Model connects all three system layers into a single forward pass:

Component	Description
Layer 1 — Backbone	MentalRoBERTa loaded from the Phase 1 checkpoint; produces 768-dim token embeddings
Layer 2 — GAT	gat1: 768 → 4 heads × 256 = 1024 (concatenated); gat2: 1024 → 2 heads → 256 (averaged). Token embeddings are treated as graph nodes with sequential edges (token i → token i+1), cached per batch-size/sequence-length pair
Layer 3 — Emotion head	256 → 128 → 28, multi-label (sigmoid applied in loss/inference)

Component	Description
Layer 3 — MH category head	256 → 64 → 2, binary softmax classification
Layer 3 — Severity head	256 → 64 → 1, regression with sigmoid output; all parameters frozen (requires_grad = False) pending DAIC-WOZ access
Pooling	CLS-token embedding (position 0 of each sequence) taken as the graph-level vector after both GAT layers

Loading the Phase 1 checkpoint into a base AutoModel produces the expected UNEXPECTED/MISSING warnings for the old 2-class sequence-classifier head and the fresh pooler — this is correct behaviour, not an error, since Phase 1 saved a classifier and Phase 3 only needs the base transformer.

6. Transfer of Phase 2 GAT Weights

The Phase 2 GAT checkpoint (gat_phase2_best.pt) is loaded and matched against Phase 3's model by key name and shape.

Layers matched	8 / 10 (gat1 and gat2: att_src, att_dst, bias, lin.weight for each)
Layers not matched	2 — Phase 2's classifier.weight and classifier.bias (the old 9-class PRIMATE symptom head), intentionally replaced by the new three-head design

The GAT layers carry forward whatever they learned in Phase 2 about which token relationships mattered for PHQ-9 symptoms, and continue fine-tuning from there in the context of emotion classification, rather than starting from randomly initialised weights — subject to the edge-construction caveat noted in Section 2.1.

7. Loss Functions and Differential Learning Rates

Emotion loss	BCEWithLogitsLoss (multi-label)
MH category loss	CrossEntropyLoss (binary, mutually exclusive)
Combined loss	0.6 × BCE_emotion + 0.4 × CE_mh
Backbone learning rate	2e-5 (low, to preserve pretrained knowledge)
GAT + heads learning rate	5e-5 (higher, to adapt these layers faster)
Optimizer	AdamW, weight decay = 0.01
Severity head	Excluded from the optimizer entirely — parameters frozen

The 0.6/0.4 split weights emotion classification more heavily since GoEmotions is the primary training signal, while the MH label — being synthetically derived — is given less weight.

8. Training Progress (3 Epochs)

Epoch	Avg Train Loss	Val F1-macro
1	0.1326	0.4573
2	0.1144	0.4885
3	0.1067	0.5008 (best, checkpoint saved)

3,082 gradient update steps per epoch (49,307 training samples ÷ batch size 16). Emotion predictions are thresholded at 0.3 sigmoid probability rather than 0.5, since GoEmotions labels are sparse and a lower threshold improves recall on rare *emotions without a large precision cost. Loss decreases and validation F1 improves consistently across all three epochs — stable learning with no signs of overfitting in this short run — and the model crosses 0.50 macro-F1 by epoch 3, consistent with published BERT-based baselines on GoEmotions (0.46–0.54).*

9. Final Test Evaluation

The best checkpoint (epoch 3, Val F1 = 0.5008) was loaded and evaluated on the held-out test set of 4,351 samples.

9.1 Overall Metrics

Macro F1 (all 28 classes equally weighted)	0.49
Micro F1 (weighted by class frequency)	0.59
Samples F1 (per-sample average)	0.61

9.2 Per-Emotion Breakdown

Emotion	Precision	Recall	F1	Support
admiration	0.70	0.71	0.70	764
amusement	0.85	0.71	0.78	408
anger	0.52	0.54	0.53	417
annoyance	0.48	0.54	0.51	712
approval	0.51	0.48	0.50	990
caring	0.46	0.50	0.48	343
confusion	0.52	0.64	0.57	396
curiosity	0.63	0.79	0.70	432
desire	0.50	0.44	0.46	181
disappointment	0.41	0.47	0.44	490
disapproval	0.47	0.58	0.52	615
disgust	0.47	0.47	0.47	303
embarrassment	0.42	0.23	0.30	132
excitement	0.39	0.39	0.39	318
fear	0.54	0.58	0.56	167
gratitude	0.78	0.73	0.76	385
grief	1.00	0.04	0.08	49
joy	0.50	0.45	0.47	433
love	0.82	0.75	0.78	335
nervousness	0.34	0.24	0.28	119
optimism	0.50	0.48	0.49	482

Emotion	Precision	Recall	F1	Support
pride	1.00	0.01	0.02	99
realization	0.41	0.25	0.31	529
relief	0.45	0.10	0.16	92
remorse	0.81	0.47	0.60	127
sadness	0.54	0.55	0.55	365
surprise	0.55	0.47	0.51	267
neutral	0.70	0.89	0.78	2,352
Macro avg	0.58	0.48	0.49	12,302
Micro avg	0.58	0.60	0.59	12,302
Weighted avg	0.58	0.60	0.58	12,302
Samples avg	0.64	0.67	0.61	12,302

9.3 Interpretation

Performance splits cleanly by class frequency:

- **Strong (F1 > 0.70):** admiration (0.70), amusement (0.78), gratitude (0.76), love (0.78), neutral (0.78), curiosity (0.70) — high-frequency, clearly expressed emotions with plenty of training examples.
- **Moderate (F1 0.44–0.60):** anger, annoyance, approval, confusion, disapproval, fear, remorse, sadness, surprise — semantically overlapping emotions that are harder to distinguish even for human raters.
- **Low (F1 < 0.30):** grief (0.08), pride (0.02), relief (0.16), nervousness (0.28) — rare classes with very few training examples (49, 99, 92, and 119 test samples respectively). Precision on grief and pride is a perfect 1.00, meaning every prediction the model does make is correct, but recall collapses to near zero — the model almost never predicts them at all. This is the class-imbalance problem GoEmotions is known for, not a modelling error.

Micro-F1 (0.59) and samples-F1 (0.61) are both noticeably higher than macro-F1 (0.49) precisely because they weight by frequency — in practice, most predictions the model makes fall on the common, well-performing emotions, and the macro average is pulled down by a handful of very rare classes.

10. Key Outputs and Deliverables

Deliverable	Location	Purpose
phase3_best.pt	/kaggle/working/	Best checkpoint by validation macro-F1 (epoch 3), used for final test evaluation
phase3_final.pt	/kaggle/working/	Final saved model state after test evaluation
phase3_tokenizer/	/kaggle/working/	Tokenizer saved alongside the model for downstream reuse

11. Phase 3 Summary

Metric	Value	Notes
Training samples (GoEmotions)	49,307	85% of 58,009 deduplicated comments
Train / Val / Test split	49,307 / 4,351 / 4,351	85% / 7.5% / 7.5%
Phase 2 GAT weights transferred	8 / 10 layers	Old 9-class PRIMATE head intentionally excluded
Best validation macro-F1	0.5008	Epoch 3 of 3
Test macro-F1	0.49	Strong on common emotions, weak on rare ones (see Section 9.3)
Test micro-F1 / samples-F1	0.59 / 0.61	More representative of typical per-comment performance
Severity head	Frozen	Included in architecture; untrained pending DAIC-WOZ access

12. Next Steps — Phase 4 and Remaining Phase 3 Work

Phase 3 is functionally complete for the emotion and MH-category heads, but the severity head — and full validation of the joint model — depends on data still pending approval.

- Submit / follow up on the DAIC-WOZ data use agreement if not already approved — this blocks unfreezing and training the severity head
- Once DAIC-WOZ access is granted, unfreeze the severity head and fine-tune it jointly with the rest of the pipeline on PHQ-8 severity scores
- Revisit the sequential-edge vs. dependency-edge discrepancy noted in Section 2.1 — consider whether a differentiable or precomputed dependency-graph structure could be reintroduced without sacrificing training speed
- Investigate the near-zero recall on rare emotions (grief, pride, relief, nervousness) — e.g. targeted oversampling or a lower decision threshold specifically for low-support classes
- Replace the synthetic MH-category proxy label with a more clinically grounded signal once a suitable labelled dataset is available
- Begin scoping Phase 4 — Groq LLM integration for report generation and crisis alerting — once the full three-head model (including severity) is validated

13. Phase Status

Phase	Status	Notes
Phase 1 — Backbone Fine-Tuning	✓ COMPLETE	Checkpoint committed to Kaggle Output
Phase 2 — GAT Training	✓ COMPLETE	gat_phase2_final.pt saved; test macro-F1 0.5234
Phase 3 — Three-Head Training	IN PROGRESS	Emotion + MH heads trained (macro-F1 0.49); severity head frozen pending DAIC-WOZ

Phase	Status	Notes
Phase 4 — LLM Integration	PENDING	Groq API integration — awaiting Phase 3 completion